

Comparing Moral Reasoning in Large Language Models: Mistral and Gemma3 on the ETHICS Dataset

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July 2026

1 Introduction

Nowadays, Large Language Models (LLMs) are used to answer questions that involve ethical reasoning. These models are capable of generating coherent responses, but it remains quite unclear how they handle morally ambiguous situations and what reasoning strategies they employ. Understanding this behaviour is important because these models, developed on deep learning and natural language processing technologies, have rapidly integrated into diverse sectors, including healthcare, where they have begun to transform various aspects of medical practice [1].

This project investigates the moral reasoning capabilities of two open-source LLMs, Mistral and Gemma3, executed locally through Ollama, analysing how the models justify their answers, express uncertainty, and structure their reasoning when responding to ethically challenging scenarios. Three subsets extracted from the ETHICS dataset are used to represent different types of moral scenarios: Commonsense, Deontology, and Justice [2].

The objective is to compare their linguistic behaviour and reasoning in these ethical contexts using quantitative and qualitative evaluation metrics.

2 Research Question and Methodology

2.1 Research Question

The main research question is as follows. How do Mistral and Gemma3 Large Language Models respond to morally ambiguous scenarios, and what differences can be observed in their reasoning strategies, linguistic behaviour, execution time, and label classification performance?

2.2 Methodology

The workflow was divided into four sections.

2.2.1 Dataset Exploration and Preprocessing

The first section consisted of exploring the three above-mentioned datasets, where Commonsense data details social dilemmas, like personal relationship conflicts from online forums, requiring the model to analyse complex nuances over an extended text body.

Deontology and Justice files present direct scenarios paired with a categorical binary variable called "label" to indicate the morality of an action. The binary label for Deontology indicates whether the excuse is considered morally justified (1) or not morally justified (0), while for Justice, label 1 represents a fair justification, whereas 0 represents an unfair justification. In more detail, Deontology focuses on daily duties and excuses, like questioning whether one should make the bed, while Justice evaluates fairness in decision-making, like a teacher skipping a scheduled exam, providing a benchmark for model evaluation.

Since the original datasets contain hundreds and thousands of examples, small representative subsets were selected to reduce computational cost during experimentation: 1 example for Commonsense, 4 samples for Deontology and 4 for Justice. Although the subsets are not intended to provide statistically significant conclusions, they allow a small comparison of the reasoning behaviour of the evaluated models.

In addition, preprocessing included splitting the Commonsense raw texts into titles and stories, removing unnecessary whitespace, and replacing abbreviations AITA ("Am I the Asshole") and WIBTA ("Would I Be the Asshole") with their complete forms to improve prompt readability.

2.2.2 Prompt Generation

The prompts were generated using a module called PromptBuilder. Different prompt templates were designed for each dataset while maintaining a consistent instruction format. The prompts requested not only a moral judgment but also a brief explanation. For the Deontology and Justice samples, the prompts required the models to answer using predefined output labels (VALID/INVALID and FAIR/UNFAIR), allowing the responses to be automatically mapped to binary values and directly compared with the ground-truth labels.

2.2.3 Response Generation

Mistral and Gemma3 were executed locally through Ollama using a common interface implemented in the module called LLMInterface. The same prompts were submitted to both models under identical conditions. Execution time was measured for every dataset to compare computational efficiency.

2.2.4 Evaluation

The generated responses were evaluated using quantitative and qualitative metrics. Quantitative evaluation included: Label classification accuracy (only for

Deontology and Justice), average response length (measures the average number of words generated by the language model for each prompt), refusal rate (measures the percentage of responses in which the language model refuses to answer a prompt; it is computed by detecting "I cannot", "I can't help", or "cannot assist"), and total execution time.

Qualitative evaluation was performed using a classifier based on the presence of some words describing judgment (like "right", "wrong", "acceptable"), justification (like "because", "since", "therefore"), uncertainty (like "may", "might", "could"), and reframing strategies (responses that contain: "it is important to consider", "from a moral perspective", "on the other hand", "in this case").

The complete implementation follows a modular architecture composed of separate Python modules for dataset loading, prompt generation, model interaction, response analysis, and evaluation, while Jupyter notebooks were used for experimentation, comparison and visualisation.

3 Experimental results

Three experiments were conducted, one for each ethical dataset, using identical prompts for both language models. All experiments were executed locally through Ollama to guarantee reproducibility under the same computational environment.

Table 1: Comparison of evaluation metrics between Mistral and Gemma3 across the three ETHICS datasets

Metric	Mistral			Gemma3		
	Common	Deon.	Justice	Common	Deon.	Justice
Accuracy	–	100.00	50.00	–	75.00	75.00
Average length	136.00	51.25	40.00	505.00	26.50	25.25
Execution time (s)	82.93	101.72	79.87	115.64	48.86	36.90
Judgment	100.00	100.00	100.00	100.00	100.00	100.00
Justification	100.00	100.00	75.00	100.00	100.00	100.00
Reframing	0.00	0.00	0.00	0.00	0.00	0.00
Refusal rate	0.00	0.00	0.00	0.00	0.00	0.00
Uncertainty	100.00	25.00	25.00	100.00	0.00	0.00

In the Commonsense part, both models produced moral judgments and detailed explanations, where Gemma3 generated much longer responses than Mistral, with an average length of 505 against 136, but required much more execution time of 115.64 against 82.93 seconds. For the qualitative metrics, both models always expressed moral judgments and justifications in their responses, indicating that they generally explained their decisions instead of simply providing conclusions. Expressions of uncertainty also appeared in both models, suggesting a greater tendency to acknowledge ambiguity, while reframing strategies were uncommon in both models. Finally, the refusal rate was zero for both models.

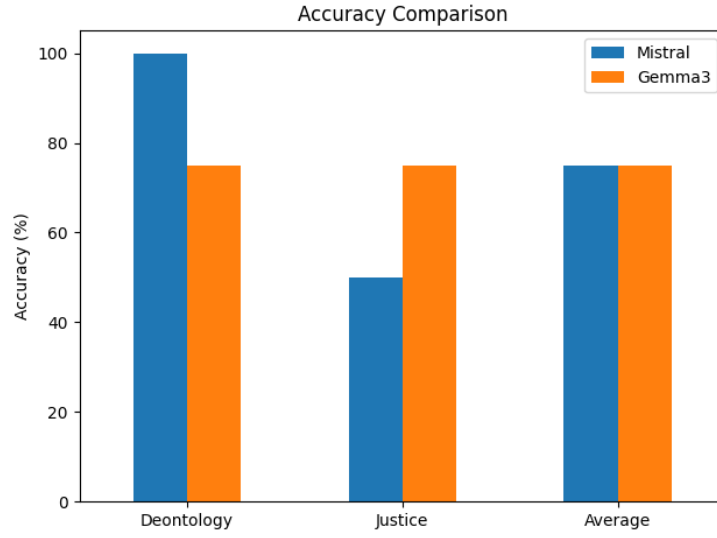


Figure 1: Accuracy label classification between Mistral and Gemma3

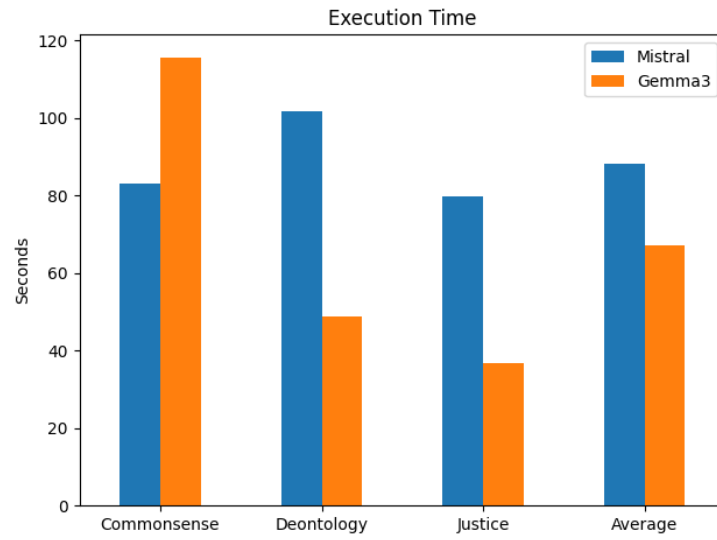


Figure 2: Computational execution time between Mistral and Gemma3

In the deontology part, Mistral achieved perfect classification accuracy, while Gemma3 got 75 percent, but did the task more quickly (48.86 against 101.72 seconds); also, the average length for Mistral was almost double that of Gemma3. While the judgment, justification, reframing and refusal rate indicators were the same as the Commonsense section, the uncertainty for Mistral was present in one out of four responses and zero appearances for Gemma3.

In the justice part, Mistral achieved 50 percent on classification accuracy, while Gemma3 got 3 out of 4 correct classification label, and completed the task more quickly (36.90 against 79.87 seconds); furthermore, the average length was higher for Mistral (40 vs. 25.25). While the judgment, reframing and refusal rate metrics were the same as the Deontology and Commonsense section, justification appears in all responses for Gemma3, and in 3 out of 4 for Mistral. Finally, uncertainty was the same as deontology part.

4 Concluding remarks

The experiments showed that Mistral and Gemma3 consistently produced moral judgments and justifications while exhibiting no refusal behaviour across the sample scenarios. Mistral achieved higher classification accuracy on the Deontology dataset, while Gemma3 performed better on the Justice dataset and required less execution time for the structured classification tasks. Regarding the response generated in Commonsense, Gemma3 involved much more context, leading to more terms, analysis, and study of the situation, and ended the response by generating questions for the user to continue the conversation, while Mistral was more direct, but comparatively, both reach the same outcome, which was that the action taken by the person in the sample is not morally acceptable. Although the experiments were conducted on very small dataset samples due to limit computational cost, the proposed methodology proved effective for comparing qualitative and quantitative aspects of ethical reasoning. Future work could extend the evaluation of a higher number of samples or even the complete ETHICS datasets, include additional language models and prompting strategies, and incorporate more advanced semantic evaluation techniques.

AI Usage Disclaimer This project was developed with the assistance of OpenAI’s ChatGPT (GPT-5). The AI was used to support the design of the experimental methodology, the implementation of Python code, the development of prompt templates, the organisation of the project structure, the preparation of evaluation metrics and visualisations, part of the drafting and revision of descriptive texts, and part of the interpretation of experimental results. AI-generated outputs were reviewed, tested, adapted, and integrated into the final project by the author. All code was executed and experimentally validated, all generated texts were revised and edited before inclusion, and the final decisions regarding the methodology, implementation, analysis, and conclusions were made by the author, who takes full responsibility for the correctness, accuracy, and academic integrity of the submitted work.

References

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